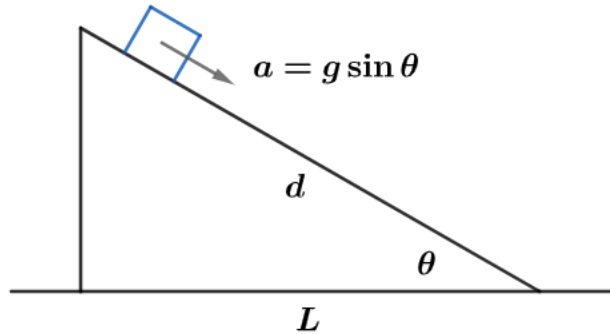


2021 F=ma Exam: Problem 15

Kevin S. Huang



From dynamics, the block has an acceleration $a = g \sin \theta$ down the plane which has length $d = L / \cos \theta$. From kinematics,

$$\begin{aligned} d &= \frac{1}{2}at^2 \\ \frac{L}{\cos \theta} &= \frac{1}{2}gt^2 \sin \theta \\ t^2 &\propto \frac{1}{\sin \theta \cos \theta} = \frac{2}{\sin(2\theta)} \end{aligned}$$

Thus, to minimize t , we maximize $\sin(2\theta)$ to obtain $\theta = \frac{\pi}{4}$ so the answer is B.